

Chapter 30: Best Approximation Algorithms

Convex Analysis and Monotone Operator Theory

Lecture Notes

July 14, 2026

Overview

- 1 Introduction and Halper's Algorithm
- 2 Dykstra's Algorithm
- 3 Haugazeau's Algorithm
- 4 Applications and Numerical Examples
- 5 Key Theoretical Results
- 6 Exercises and Further Reading

Best Approximation Problem

Core Problem

Given a nonempty closed convex subset C of a Hilbert space $\mathcal{H}$ and a point $x \in \mathcal{H}$, find the unique point $p \in C$ that minimizes $\|x - y\|$ over all $y \in C$.

Definition

For a point x , the **projection** onto C is

$$P_C x := \arg \min_{y \in C} \|x - y\|$$

- Projections exist and are unique (by Hilbert space theory)
- P_C is a nonexpansive operator: $\|P_C x - P_C y\| \leq \|x - y\|$
- Main goal: Design iterative algorithms converging to $P_C x$

Halper's Algorithm

Theorem (Halper's Algorithm)

Let D be a nonempty closed convex subset of $\mathcal{H}$, let $T : D \rightarrow D$ be a nonexpansive operator such that $C = \text{Fix } T \neq \emptyset$, and let $(\lambda_n)_n$ be a sequence in $[0, 1]$ such that $\lambda_n \rightarrow 0$, $\sum_n \lambda_n = +\infty$, and $\sum_n |\lambda_{n+1} - \lambda_n| < +\infty$. Let $x_0 \in D$ and set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = \lambda_n x + (1 - \lambda_n) T x_n.$$

Then $x_n \rightarrow P_{\text{Fix } T} x$.

Key convergence conditions:

- ① Decreasing step sizes: $\lambda_n \rightarrow 0$
- ② Summability: $\sum_n \lambda_n = +\infty$
- ③ Uniform boundedness: $\sum_n |\lambda_{n+1} - \lambda_n| < +\infty$

Halper's Algorithm: Python Implementation

```
def halpers_algorithm(T, x, x0, lambda_seq, max_iter=1000, tol=1e-8):  
    """  
    Halper's fixed point algorithm.  
  
    Parameters:  
    -----  
    T : function  
        Nonexpansive operator (fixed point operator)  
    x : array  
        Target point (for projection from fixed points)  
    x0 : array  
        Initial point  
    lambda_seq : list  
        Step sizes (decreasing to 0)  
    """  
    x_curr = np.array(x0, dtype=float)  
    history = [x_curr.copy()]  
  
    for n in range(min(max_iter, len(lambda_seq))):  
        lam = lambda_seq[n]  
        Tx = T(x_curr)  
        x_next = lam * x + (1 - lam) * Tx  
  
        if np.linalg.norm(x_next - x_curr) < tol:  
            break  
  
    x_curr = x_next
```

Corollary 30.2: Finite Family of Operators

Corollary

Let D be a nonempty closed convex subset, let $(T_i)_{i \in I}$ be a finite family of nonexpansive operators from D to D such that $C = \bigcap_{i \in I} \text{Fix } T_i \neq \emptyset$. Let $x \in D$ and $(\omega_i)_{i \in I}$ be real numbers in $[0, 1]$ with $\sum_{i \in I} \omega_i = 1$. Let $(\lambda_n)_n$ be a sequence satisfying the conditions of Theorem 30.1. Let $x_0 \in D$ and set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = \lambda_n x + (1 - \lambda_n) \sum_{i \in I} \omega_i T_i x_n.$$

Then $x_n \rightarrow P_C x$.

This is useful for finding projections onto intersections of convex sets!

Corollary 30.3: Averaged Nonexpansive Operators

Corollary

Let D be a nonempty closed convex subset, let m be a strictly positive integer, set $I = \{1, \dots, m\}$, let $(T_i)_{i \in I}$ be a family of averaged nonexpansive operators from D to D such that $C = \bigcap_{i \in I} \text{Fix } T_i \neq \emptyset$. Let $x \in D$ and $(\lambda_n)_n$ be a sequence satisfying Theorem 30.1 conditions. Let $x_0 \in D$ and set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = \lambda_n x + (1 - \lambda_n) T_1 \cdots T_m x_n.$$

Then $x_n \rightarrow P_C x$.

Example (Application: Cyclic projections)

When each $T_i = P_{C_i}$ is a projection, the composite operator $P_{C_1} \cdots P_{C_m}$ achieves strong convergence to $P_C x$ where $C = \bigcap_i C_i$.

Linear Mean Ergodic Theorem

Corollary (30.5: Linear Mean Ergodic Theorem)

Let $T \in \mathcal{B}(\mathcal{H})$ be nonexpansive and let $x \in \mathcal{H}$. Then

$$\frac{1}{n+1} \sum_{k=0}^n T^k x \rightarrow P_{\ker(I-T)} x.$$

Key insight: The Cesàro average of iterates converges to the projection onto the kernel of $(I - T)$, which is the set of fixed points of T .

- This provides strong convergence without requiring decreasing step sizes
- The convergence is in the mean (averaging)
- Weaker than pointwise convergence but easier to achieve

Dijkstra's Algorithm: Motivation

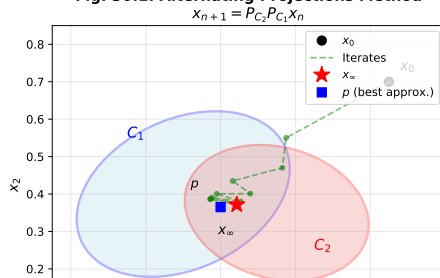
Problem

Find the projection of a point $x_0 \in \mathcal{H}$ onto the intersection of m closed convex sets $C_1, \dots, C_m$.

Key challenges:

- 1 Computing $P_{C_1 \cap \dots \cap C_m}$ directly is often difficult
- 2 But projections P_{C_i} onto individual sets may be easy to compute
- 3 Dykstra's algorithm combines individual projections

Fig. 30.1: Alternating Projections Method



Theorem 30.7: Dykstra's Algorithm

Theorem (Dykstra's Algorithm)

Let m be a strictly positive integer, set $I = \{1, \dots, m\}$, let $(C_i)_{i \in I}$ be a family of closed convex subsets of $\mathcal{H}$ such that $C = \bigcap_{i \in I} C_i \neq \emptyset$, and let $x_0 \in \mathcal{H}$. Set

$$i : \mathbb{N} \rightarrow I : n \mapsto 1 + \text{rem}(n - 1, m), \quad (1)$$

where $\text{rem}(\cdot, m)$ is the remainder of division by m . For every strictly positive integer n , set $P_n = P_{C_{i(n)}}$ where $C_n = C_{i(n)}$ if $n > m$. Moreover, set $q_{-(m-1)} = \dots = q_{-1} = q_0 = 0$ and

$$\text{for } n = 1, 2, \dots \quad \begin{cases} x_n = P_n(x_{n-1} + q_{n-m}) \\ q_n = x_{n-1} + q_{n-m} - x_n \end{cases} \quad (2)$$

Then $x_n \rightarrow P_C x_0$.

Dykstra's Algorithm: Understanding the Recursion

The key variables are:

- x_n : the iterate (approximation to the projection)
- q_n : a residual/correction term

The update can be understood as:

$$x_n = P_{C_i(n)}(x_{n-1} + q_{n-m}) \quad (3)$$

This means:

- 1 Add the accumulated correction from m iterations ago
- 2 Project onto the current set
- 3 Record the projection residual as $q_n = (x_{n-1} + q_{n-m}) - x_n$

The corrections q_n help the algorithm converge faster than simple alternating projections.

Dykstra's Algorithm: Python Implementation

```
def dykstra_algorithm(projections, x0, m, max_iter=1000, tol=1e-8):  
    """Dykstra's algorithm for projection onto intersection of m sets."""  
    x = np.array(x0, dtype=float)  
    q = [np.zeros_like(x) for _ in range(m)]  
    history = [x.copy()]  
  
    for n in range(1, max_iter + 1):  
        # Index of current set (cyclic)  
        i = (n - 1) % m  
  
        # Dykstra step  
        x_old = x.copy()  
        if n <= m:  
            correction = np.zeros_like(x)  
        else:  
            correction = q[(i - m) % m]  
  
        x = projections[i](x + correction)  
        q[i] = (x_old + correction) - x  
  
        history.append(x.copy())  
        if np.linalg.norm(x - x_old) < tol:  
            break  
  
    return x, np.array(history)
```

Corollary 30.4: Linear Mean Convergence for Dykstra

Corollary (Application to Two Sets)

Consider the application of Dykstra's algorithm (Theorem 30.7) to $m = 2$ intersecting closed convex sets C and D of $\mathcal{H}$.

$$\begin{cases} y_n = P_C(x_n + p_n) \\ p_{n+1} = x_n + p_n - y_n \\ x_{n+1} = P_D(y_n + q_n) \\ q_{n+1} = y_n + q_n - x_{n+1} \end{cases}$$

with $p_0 = q_0 = 0$. Then $x_n \rightarrow P_{C \cap D} x_0$ linearly.

Advantages over alternating projections:

- 1 **Linear convergence** (not just weak convergence)
- 2 **Exact** when sets are affine subspaces
- 3 More efficient for practical problems

Haugazeau's Algorithm: Motivation

Extension of Dykstra

Haugazeau's algorithm generalizes Dykstra's approach to handle:

- Families of **firmly quasinonexpansive** operators (more general than projections)
- Cyclic application of operators
- Non-cyclic scheduling

Application areas:

- 1 Projection onto intersections of convex sets
- 2 Minimizing sums of convex functions
- 3 Solving operator equations and inclusions

Theorem 30.8: Haugazeau's Algorithm

Theorem (Haugazeau's Algorithm)

Let $(T_i)_{i \in I}$ be a finite family of firmly quasinonexpansive operators from $\mathcal{H}$ to $\mathcal{H}$ such that the operators $(Id - T_i)_{i \in I}$ are demiclosed at 0 and $C = \bigcap_{i \in I} \text{Fix } T_i \neq \emptyset$. Let $x_0 \in \mathcal{H}$. Let $i : \mathbb{N} \rightarrow I$ be such that there exists a strictly positive integer m for which

$$(\forall i \in I)(\forall n \in \mathbb{N}) \quad i \in \{i(n), \dots, i(n + m - 1)\},$$

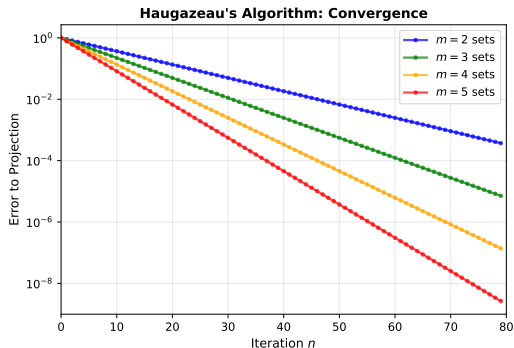
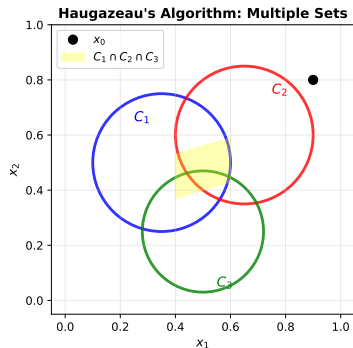
and set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = Q(x_0, x_n, T_{i(n)}x_n),$$

where Q is defined in Theorem 29.39. Then $x_n \rightarrow P_C x_0$.

Key assumption: **Cyclic scheduling** - each operator is applied regularly (at least once every m iterations).

Haugazeau's Algorithm: Visualization



The algorithm handles multiple sets and converges to their intersection.

Corollary 30.9: Finite Family of Continuous Convex Functions

Corollary

Let $(f_i)_{i \in I}$ be a finite family of continuous convex functions from $\mathcal{H}$ to $\mathbb{R}$ such that $C = \bigcap_{i \in I} \text{lev}_{\leq 0} f_i \neq \emptyset$ and suppose that one of the following holds:

- (i) The functions $(f_i)_{i \in I}$ are bounded on every bounded subset of $\mathcal{H}$.
- (ii) The functions $(f_i)_{i \in I}^*$ are supercoercive.
- (iii) $\mathcal{H}$ is finite-dimensional.

Then $x_n \rightarrow P_C x_0$ via Algorithm 30.64.

This applies Haugazeau's algorithm to feasibility problems for systems of convex inequalities.

Corollary 30.10: Firmly Nonexpansive Operators

Corollary

Let $(T_i)_{i \in I}$ be a finite family of firmly nonexpansive operators from $\mathcal{H}$ to $\mathcal{H}$ such that $C = \bigcap_{i \in I} \text{Fix } T_i \neq \emptyset$. Let $x_0 \in \mathcal{H}$, let $i : \mathbb{N} \rightarrow I$ be such that there exists a strictly positive integer m for which

$$(\forall i \in I)(\forall n \in \mathbb{N}) \quad i \in \{i(n), \dots, i(n+m-1)\},$$

and set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = Q(x_0, x_n, T_{i(n)}x_n),$$

where Q is defined in (29.39). Then $x_n \rightarrow P_C x_0$.

Note: Firmly nonexpansive operators include projections onto convex sets.

Corollary 30.11: Proximal Point Algorithm for Maximally Monotone Operators

Corollary

Let $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ be maximally monotone and such that $0 \in Ax_0$ and let $x_0 \in \mathcal{H}$. Let $J_A = (Id + A)^{-1}$ (the resolvent). Set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = Q(x_0, x_n, J_A x_n),$$

where Q is defined in (29.39). Then $x_n \rightarrow P_{\text{zer } A} x_0$.

Key result: The proximal point algorithm finds zeros of maximally monotone operators with strong convergence.

Applications: Subgradient methods, splitting algorithms, monotone inclusion problems.

Corollary 30.12: Forward-Backward Algorithm

Corollary

Let $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ be maximally monotone, let $\beta \in \mathbb{R}_{++}$, let $B : \mathcal{H} \rightarrow \mathcal{H}$ be β -cocoercive, and let $\gamma \in [0, 2\beta]$. Suppose that $\text{zer}(A + B) \neq \emptyset$. Let $x_0 \in \mathcal{H}$ and set

$$\text{for } n = 0, 1, \dots \quad \begin{cases} y_n = x_n - \gamma Bx_n \\ z_n = (1/2)(x_n + J_{\lambda A}y_n) \\ x_{n+1} = Q(x_0, x_n, z_n) \end{cases} \quad (4)$$

where Q is defined in (29.39). Then $x_n \rightarrow P_{\text{zer}(A+B)}x_0$.

- Handles sum of maximally monotone and cocoercive operators
- Splits the computation of resolvents

Example 30.13: Composite Optimization

Example

Let $f \in \Gamma_0(\mathcal{H})$, let $\beta \in \mathbb{R}_{++}$, let $g : \mathcal{H} \rightarrow \mathbb{R}$ be convex and differentiable with $1/\beta$ -Lipschitz continuous gradient, and let $\gamma \in [0, 2\beta]$. Furthermore, suppose that $\text{Argmin}(f + g) \neq \emptyset$. Let $x_0 \in \mathcal{H}$ and set

$$\text{for } n = 0, 1, \dots \quad \begin{cases} y_n = x_n - \gamma \nabla g(x_n) \\ z_n = (1/2)(x_n + \text{Prox}_f y_n) \\ x_{n+1} = Q(x_0, x_n, z_n) \end{cases} \quad (5)$$

where Q is defined in (29.39). Then $x_n \rightarrow P_{\text{Argmin}(f+g)}x_0$.

Interpretation: Forward-backward splitting for minimizing sum of smooth and nonsmooth convex functions.

Summary: Algorithm Comparison

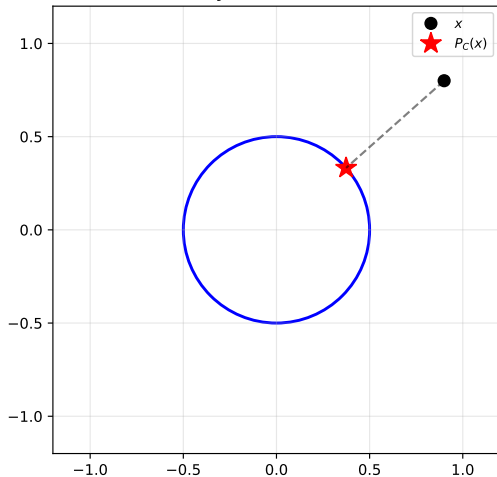
Algorithm	Convergence	Operators	Complexity
Halper's	Weak	Any nonexp.	Low
Dykstra's	Linear	Projections	Low
Haugazeau's	Strong	Firmly quasi-nonexp.	Medium
Proximal Pt.	Strong	Maximally monotone	High
Forward-Backward	Strong	$A + B$ composite	Medium

Convergence hierarchy: Weak | Linear | Strong convergence

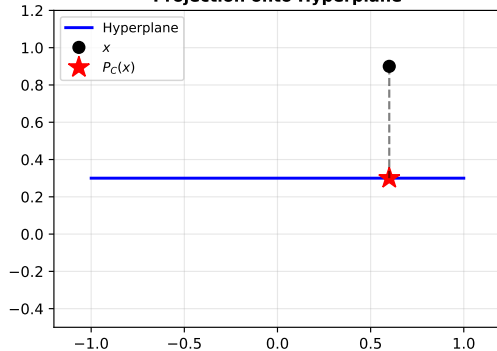
Projections onto Common Convex Sets

Examples of Projection onto Convex Sets

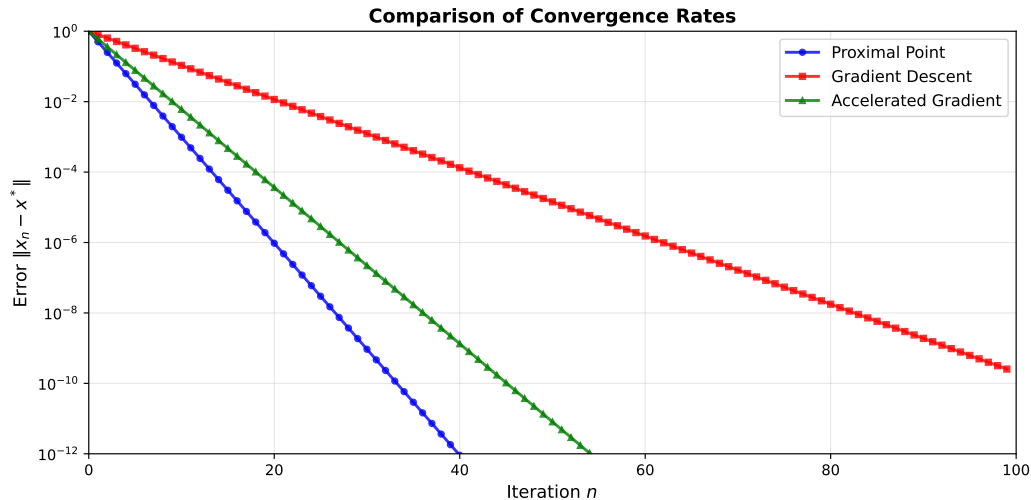
Projection onto Ball



Projection onto Hyperplane



Numerical Convergence Rates



Comparison of standard first-order methods for convex optimization.

Example: Projection onto Intersection of Two Disks

```
import numpy as np

# Define two circular sets
C1_center, C1_radius = np.array([0.3, 0.5]), 0.3
C2_center, C2_radius = np.array([0.7, 0.5]), 0.3

def proj_C1(x):
    d = np.linalg.norm(x - C1_center)
    if d <= C1_radius:
        return x
    return C1_center + C1_radius * (x - C1_center) / d

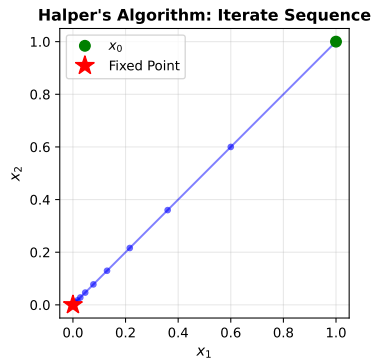
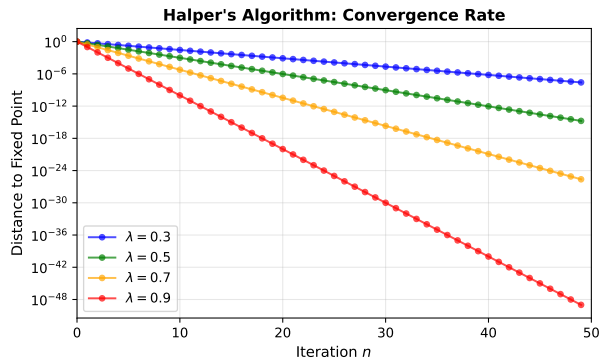
def proj_C2(x):
    d = np.linalg.norm(x - C2_center)
    if d <= C2_radius:
        return x
    return C2_center + C2_radius * (x - C2_center) / d

# Initial point
x0 = np.array([0.95, 0.7])

# Dykstra's algorithm
m = 2
projections = [proj_C1, proj_C2]
x_final, history = dykstra_algorithm(projections, x0, m)

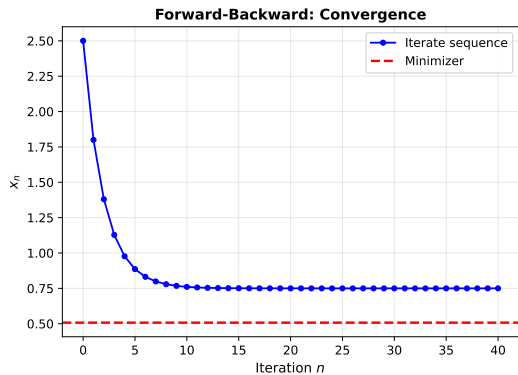
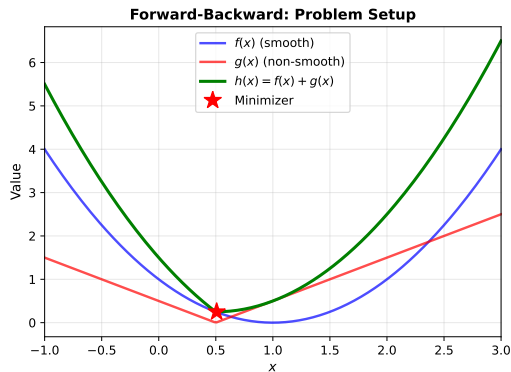
print(f"Initial point: {x0}")
```

Convergence Illustration



Left: Theoretical convergence rates depend on the parameter λ . Right: Iterates converge to the fixed point.

Forward-Backward Splitting Example



Left: Problem setup (smooth + nonsmooth objective). Right: Convergence of iterates to minimizer.

Demiclosedness and Quasinonexpansiveness

Definition (Demiclosed at 0)

An operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is **demiclosed at 0** if whenever $(x_n)_n$ is a sequence with $x_n \rightarrow x$ weakly and $(I - T)x_n \rightarrow 0$ strongly, then $(I - T)x = 0$.

Definition (Firmly Quasinonexpansive)

An operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is **firmly quasinonexpansive** if

$$\text{Fix } T = \{x : Tx = x\} \neq \emptyset \quad \text{and} \quad \|Tx - Ty\|^2 + \|Tx - y\|^2 \leq \|x - y\|^2, \quad \forall x, y \in \mathcal{H}. \quad (6)$$

Key fact: Projections are firmly nonexpansive and demiclosed.

Lemmata for Convergence Analysis

Lemma (30.6: Cauchy-Schwarz for Sequences)

Let $(\rho_n)_n \in \ell_2^2(\mathbb{N})$, let $m \in \mathbb{N}$, and set $(\forall n \in \mathbb{N}) \sigma_n = \sum_{k=0}^n \rho_k$. Then

$$\lim \sigma_n(\sigma_n - \sigma_{n-m-1}) = 0.$$

This is a key tool for analyzing the residuals in Dykstra's algorithm.

Lemma

The residual terms q_n in Dykstra's algorithm satisfy:

- They are bounded: $\sum_k \|q_k\|^2 < \infty$
- They converge to zero: $q_n \rightarrow 0$
- This ensures convergence to the fixed point

Remark 30.14: Connection to Fixed Point Methods

Remark

The principle employed in the proof of Corollary 30.12 is general:

Let $(R_i)_{i \in I}$ be a finite family of nonexpansive operators from $\mathcal{H}$ to $\mathcal{H}$ such that $\bigcap_{i \in I} \text{Fix } R_i \neq \emptyset$ and set $(\forall i \in I) T_i = (1/2)(R_i + \text{Id})$.

Then $\bigcap_i \text{Fix } T_i = \bigcap_i \text{Fix } R_i$, and it follows from Proposition 4.4 that the operators $(T_i)_{i \in I}$ are firmly nonexpansive.

Thus, Theorem 30.8 can be used to find the best approximation to a point $x_0 \in \mathcal{H}$ from $\bigcap_i \text{Fix } R_i$.

This shows that firmly nonexpansive methods are a unifying framework.

Corollary 30.15: Haugazeau's Algorithm (Cyclic Version)

Corollary (Haugazeau's Algorithm - Cyclic Projections)

Let m be a strictly positive integer, set $I = \{1, \dots, m\}$, let $(C_i)_{i \in I}$ be a family of closed convex subsets of $\mathcal{H}$ such that $C = \bigcap_{i \in I} C_i \neq \emptyset$, and let $x_0 \in \mathcal{H}$. For every $i \in I$, denote by P_i the projector onto C_i . Let $\text{rem}(\cdot, m)$ be the remainder function of division by m , and set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = Q(x_0, x_n, P_{1+\text{rem}(n,m)} x_n),$$

where Q is defined in (29.39). Then $x_n \rightarrow P_C x_0$.

This is the complete cyclic projection algorithm using Haugazeau's framework.

Selected Exercises

Exercise

[30.1] Suppose that $\mathcal{H} = \mathbb{R}$, and set $D = [-1, 1]$, $x = 0$, and $x_0 = 1$. By considering $T_1 : D \rightarrow D : x \mapsto 1$ and $T_2 : D \rightarrow D : x \mapsto -x$, show that Theorem 30.1 fails when (i) or (ii) is not satisfied.

Exercise

[30.2] Give a detailed proof of Corollary 30.5 (Linear Mean Ergodic Theorem).

Exercise

[30.4(ii)] Consider the application of Dykstra's algorithm to $m = 2$ intersecting closed convex subsets C and D of $\mathbb{R}^2$. With $H = \mathbb{R}^2$, $x_0 = (2, 2)$, $C = \{(\xi_1, \xi_2) \in \mathcal{H} \mid \xi_2 \leq 0\}$, $D = \{(\xi_1, \xi_2) \in \mathcal{H} \mid \xi_1 + \xi_2 \leq 0\}$. Compute the values of x_n for $n \in \{0, \dots, 5\}$ and draw a picture.

Key Takeaways

- 1 **Halper's Algorithm:** Solves the best approximation problem using convex combinations of operator iterates with decaying step sizes.
- 2 **Dykstra's Algorithm:** Specialized for projections onto intersections of convex sets; uses residual corrections for linear convergence.
- 3 **Haugazeau's Algorithm:** General framework for firmly quasinonexpansive operators; includes proximal point and forward-backward splitting.
- 4 **Convergence Theory:** Relies on demiclosedness, weak convergence subsequences, and properties of monotone operators.
- 5 **Applications:** Feasibility, convex optimization, monotone inclusions, machine learning algorithms.

- Bauschke, H. H., & Combettes, P. L. (2017). *Convex Analysis and Monotone Operator Theory in Hilbert Spaces* (2nd ed.). Springer.
- Chapter 30: Best Approximation Algorithms (pp. 991–1032)
 - Sections 30.1–30.3: Halper's, Dykstra's, and Haugazeau's algorithms
 - Applications to projections, monotone inclusions, and convex optimization
- Related chapters:
 - Chapter 4: Convexity, monotonicity, and firm nonexpansivity
 - Chapter 5: Convergence of sequences
 - Chapter 29: Monotone operators and resolvents

Summary of Theorems and Corollaries I

- **Theorem 30.1:** Halper's Algorithm - basic convergence
- **Corollary 30.2:** Weighted combination of operators
- **Corollary 30.3:** Composite averaged nonexpansive operators
- **Corollary 30.5:** Linear Mean Ergodic Theorem
- **Theorem 30.7:** Dykstra's Algorithm - cyclic projections
- **Theorem 30.8:** Haugazeau's Algorithm - general framework
- **Corollary 30.9:** Finite family of convex functions
- **Corollary 30.10:** Firmly nonexpansive operators
- **Corollary 30.11:** Proximal Point Algorithm
- **Corollary 30.12:** Forward-Backward Splitting
- **Corollary 30.15:** Haugazeau's Algorithm (Cyclic)

All algorithms share common features:

- Iteration based on simple operations (projections, resolvents)
- Convergence guaranteed under appropriate conditions
- Applicable to diverse optimization and operator problems